Ass10_32

March 24, 2026

[1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

[2]:
```
from sklearn.datasets import load_iris

iris = load_iris()

df = pd.DataFrame(data=iris.data, columns=iris.feature_names)
df['species'] = iris.target

# Map numeric labels to names
df['species'] = df['species'].map(dict(enumerate(iris.target_names)))

df.head()
```

[2]:
```
   sepal length (cm)  sepal width (cm)  petal length (cm)  petal width (cm)  \
0                5.1               3.5                1.4               0.2
1                4.9               3.0                1.4               0.2
2                4.7               3.2                1.3               0.2
3                4.6               3.1                1.5               0.2
4                5.0               3.6                1.4               0.2

  species
0  setosa
1  setosa
2  setosa
3  setosa
4  setosa
```

[3]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
```

```
 0   sepal length (cm)  150 non-null    float64
 1   sepal width (cm)   150 non-null    float64
 2   petal length (cm)  150 non-null    float64
 3   petal width (cm)   150 non-null    float64
 4   species            150 non-null    object
dtypes: float64(4), object(1)
memory usage: 6.0+ KB
```

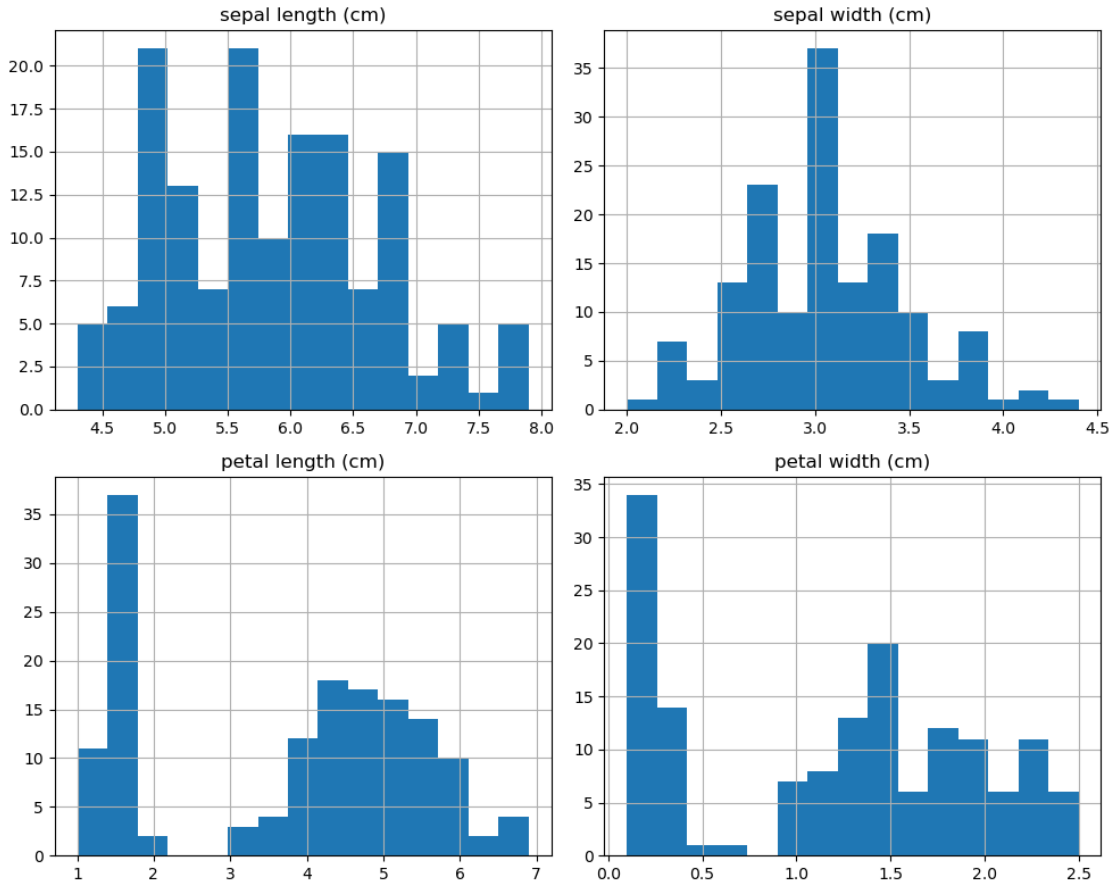
[4]:
```
feature_types = pd.DataFrame({
    "Feature": df.columns,
    "Type": df.dtypes
})

feature_types
```

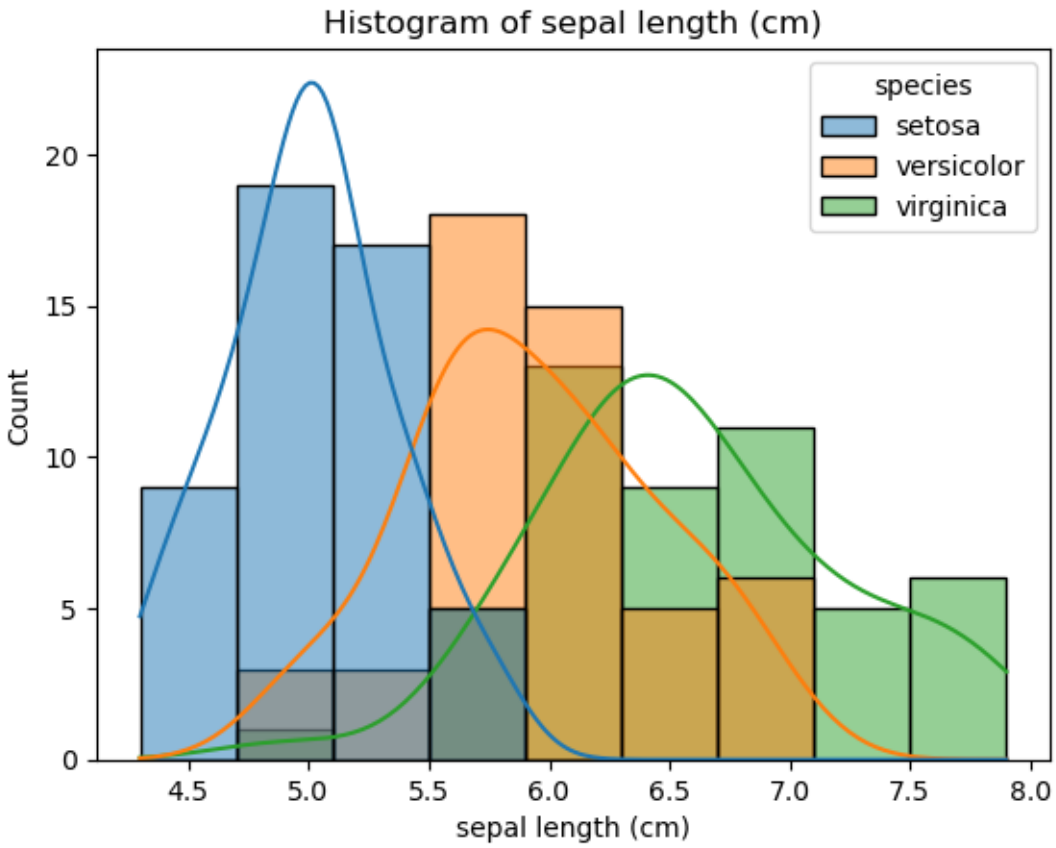
[4]:

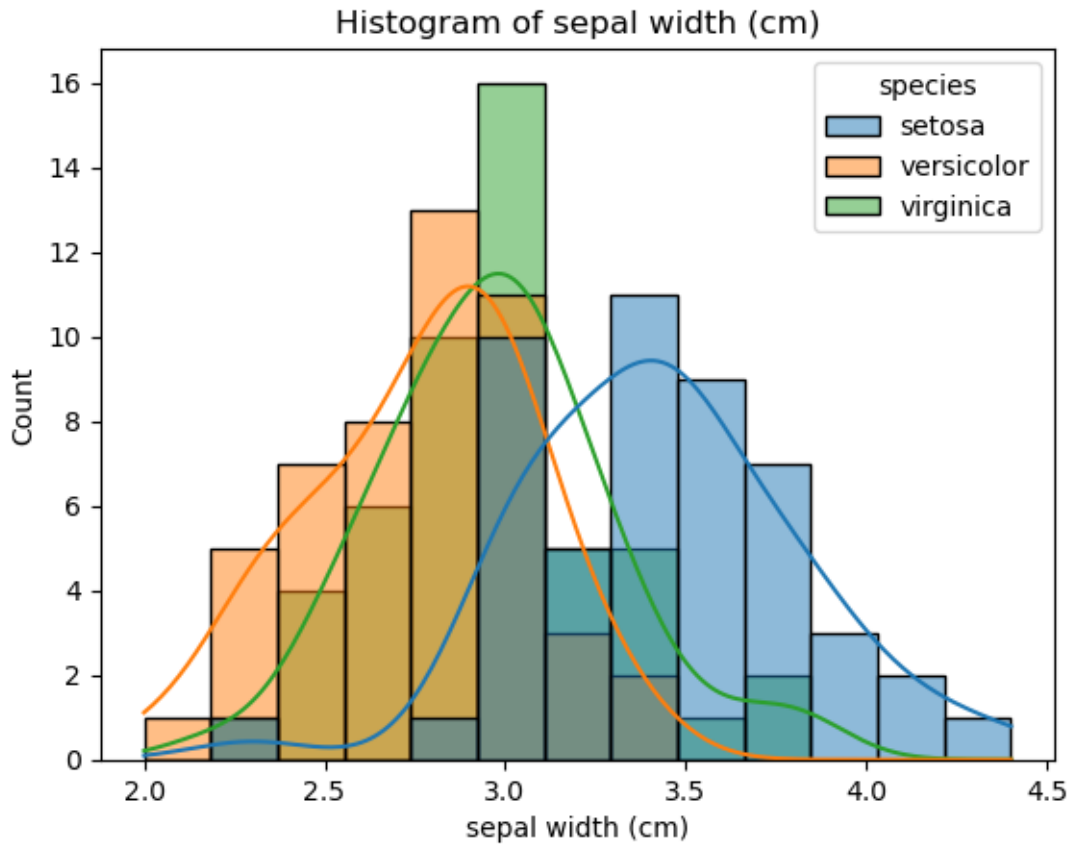
	Feature	Type
sepal length (cm)	sepal length (cm)	float64
sepal width (cm)	sepal width (cm)	float64
petal length (cm)	petal length (cm)	float64
petal width (cm)	petal width (cm)	float64
species	species	object

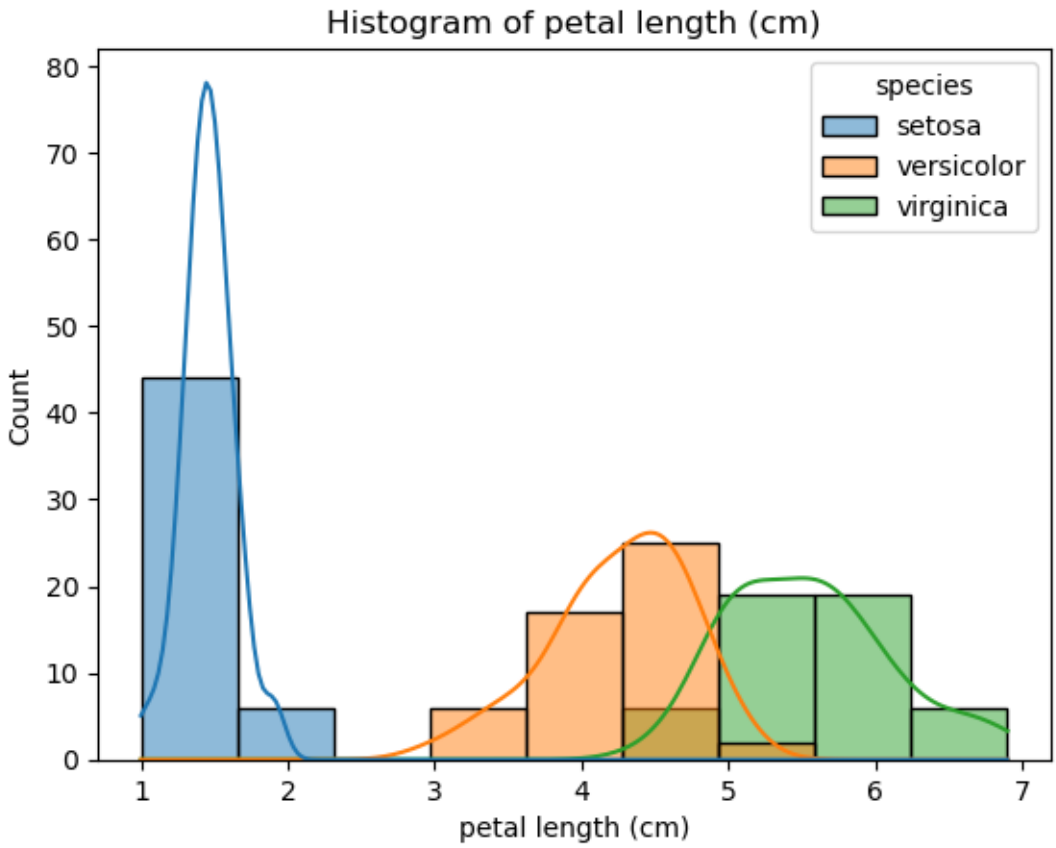
[5]:
```
df.hist(figsize=(10,8), bins=15)
plt.tight_layout()
plt.show()
```

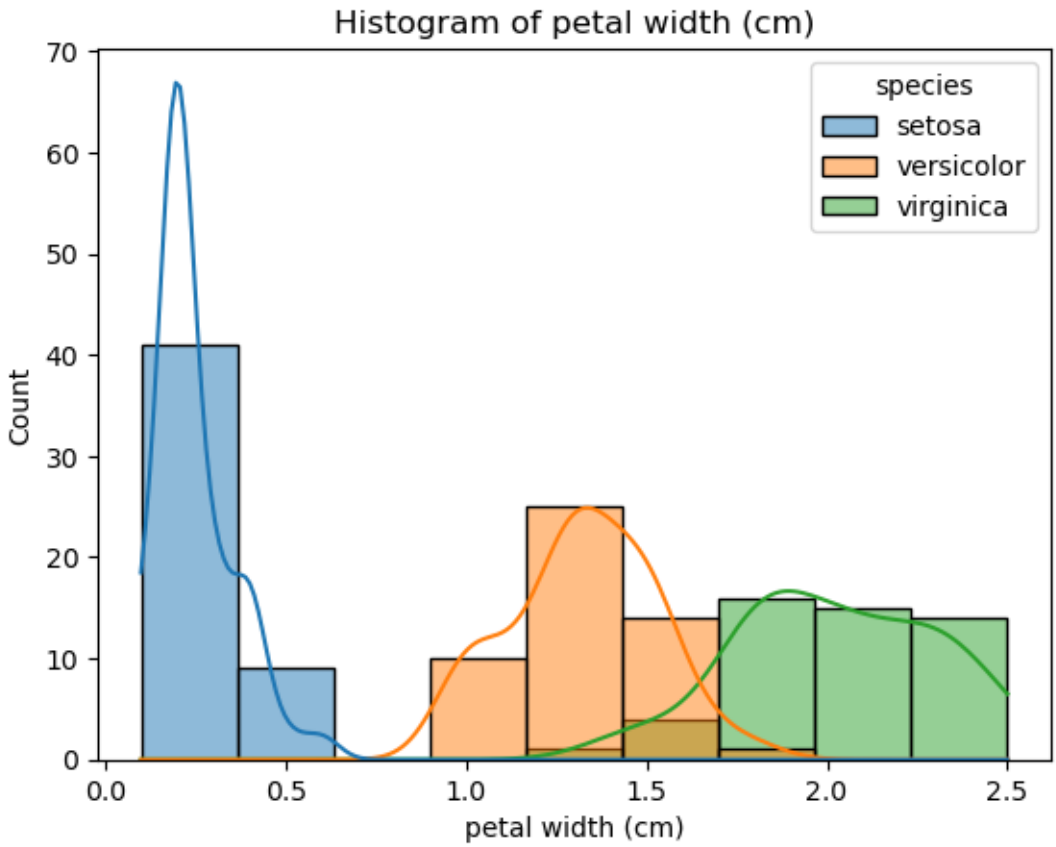


[6]:
```
for col in df.columns[:-1]:
    plt.figure()
    sns.histplot(data=df, x=col, hue='species', kde=True)
    plt.title(f"Histogram of {col}")
    plt.show()
```

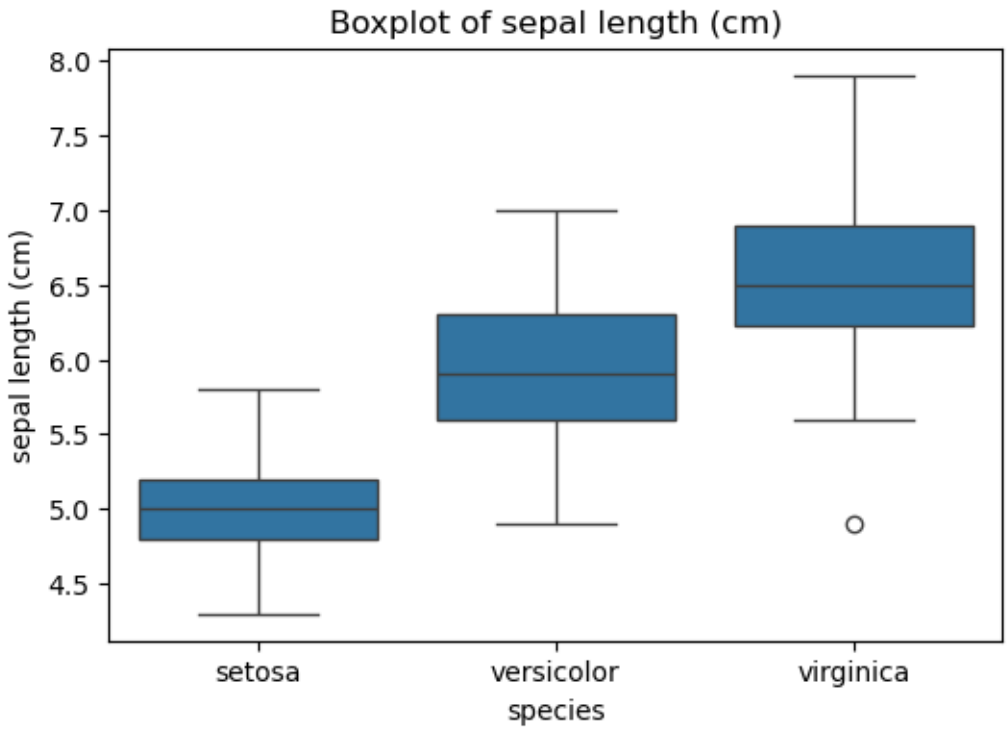

Histogram of sepal length (cm)
species
setosa
versicolor
virginica
Count
0
5
10
15
20
4.5
5.0
5.5
6.0
6.5
7.0
7.5
8.0
sepal length (cm)

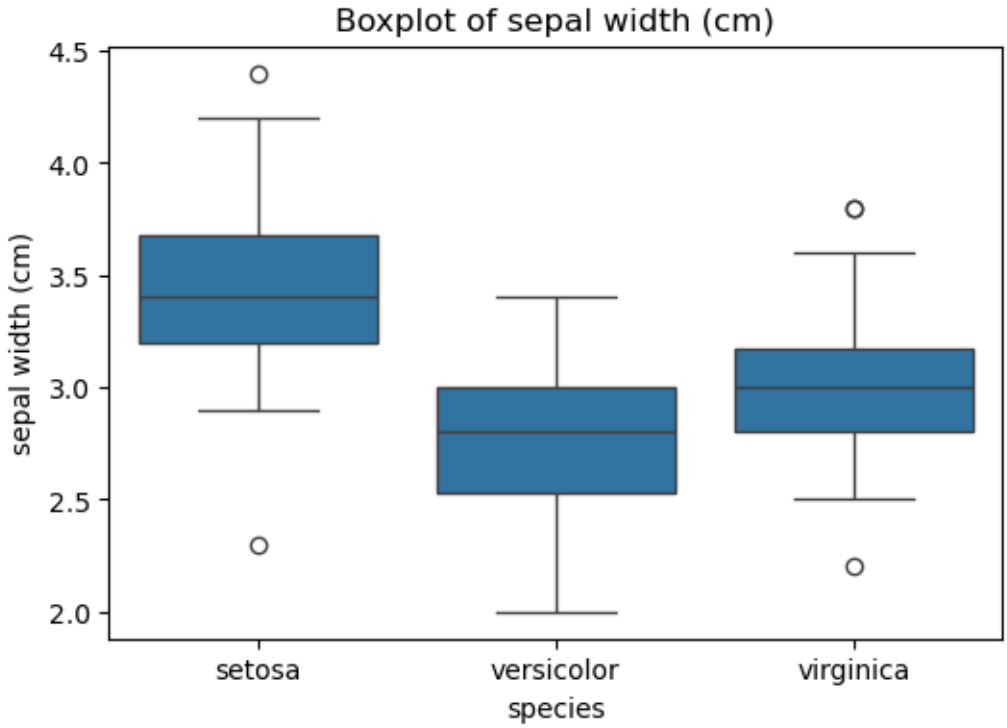

Histogram of sepal width (cm)
species
setosa
versicolor
virginica
Count
sepal width (cm)
0
2
4
6
8
10
12
14
16
2.0
2.5
3.0
3.5
4.0
4.5

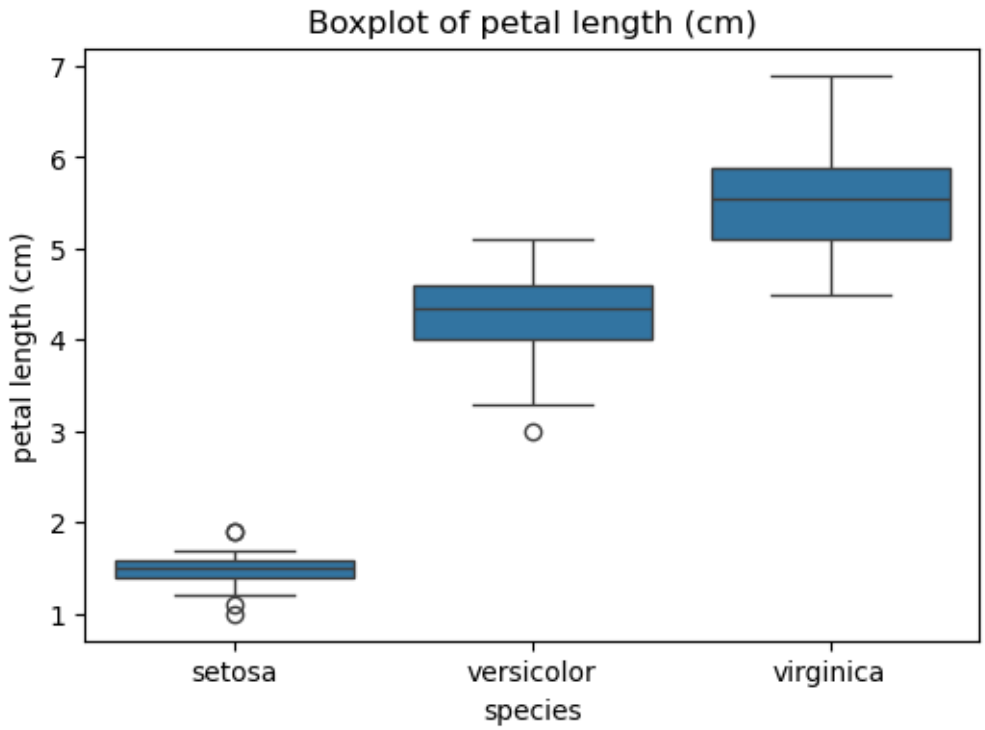

Histogram of petal length (cm)
species
setosa
versicolor
virginica
Count
petal length (cm)
0
10
20
30
40
50
60
70
80
1
2
3
4
5
6
7

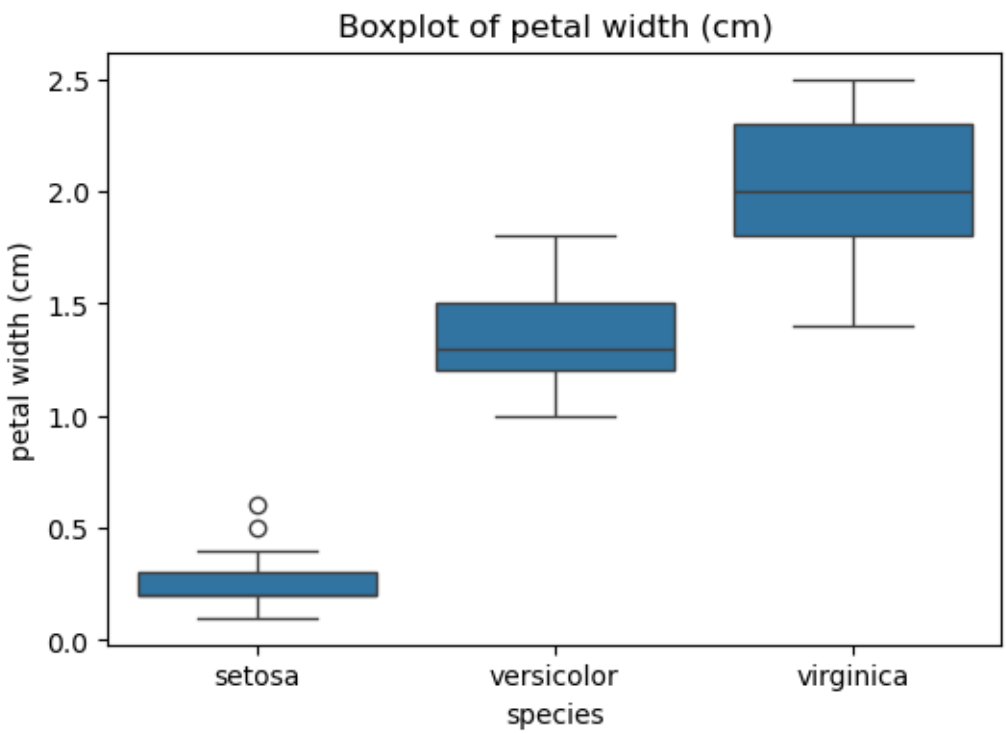


[7]:
```
for col in df.columns[:-1]:
    plt.figure(figsize=(6,4))
    sns.boxplot(x='species', y=col, data=df)
    plt.title(f"Boxplot of {col}")
    plt.show()
```

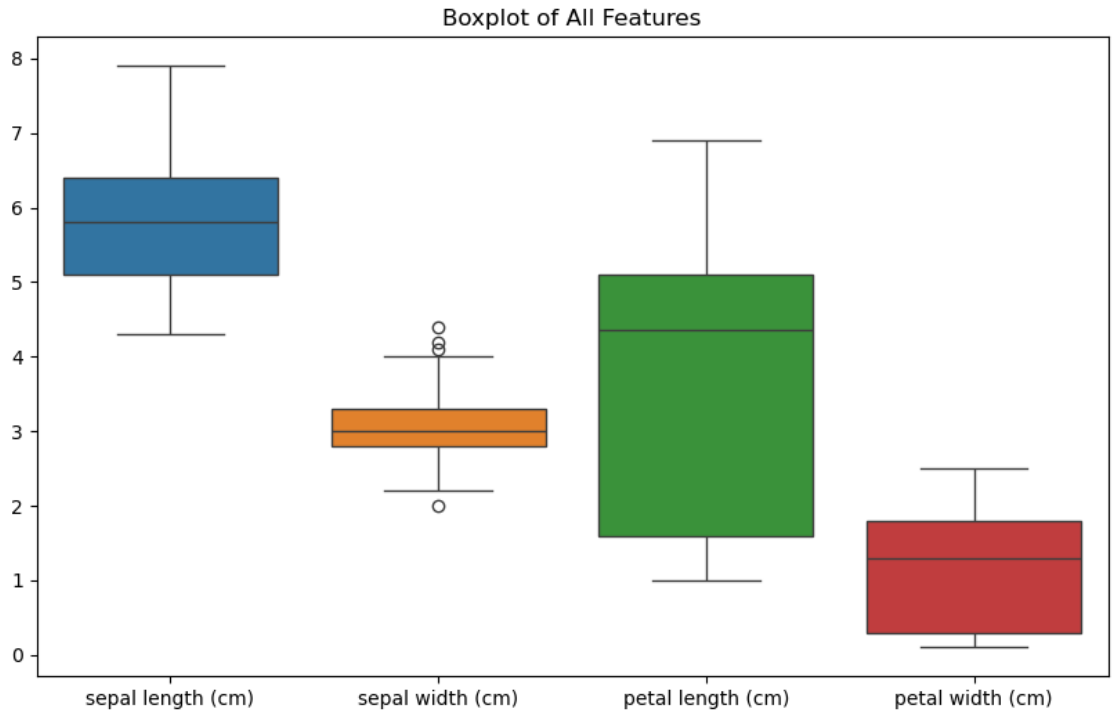

Boxplot of sepal length (cm)
sepal length (cm)
8.0
7.5
7.0
6.5
6.0
5.5
5.0
4.5
setosa
versicolor
virginica
species


Boxplot of sepal width (cm)
sepal width (cm)
4.5
4.0
3.5
3.0
2.5
2.0
setosa
versicolor
virginica
species


Boxplot of petal length (cm)
petal length (cm)
7
6
5
4
3
2
1
setosa
versicolor
virginica
species


Boxplot of petal width (cm)
petal width (cm)
2.5
2.0
1.5
1.0
0.5
0.0
setosa
versicolor
virginica
species

[8]:
```
plt.figure(figsize=(10,6))
sns.boxplot(data=df.iloc[:, :-1])
plt.title("Boxplot of All Features")
plt.show()
```



[9]:
```
df.describe()
```

[9]:
```
       sepal length (cm)  sepal width (cm)  petal length (cm)  \
count         150.000000        150.000000         150.000000
mean            5.843333          3.057333           3.758000
std             0.828066          0.435866           1.765298
min             4.300000          2.000000           1.000000
25%             5.100000          2.800000           1.600000
50%             5.800000          3.000000           4.350000
75%             6.400000          3.300000           5.100000
max             7.900000          4.400000           6.900000

       petal width (cm)
count        150.000000
mean           1.199333
std            0.762238
min            0.100000
25%            0.300000
50%            1.300000
```

```
75%            1.800000
max            2.500000
```

[ ]:

```

```